



Grampurohit Shreyas Jayant  
Electrical Engineering  
Indian Institute of Technology Bombay  
Specialization: Communication & Signal Processing

21D070029  
Dual Degree (B.Tech. + M.Tech.)  
Gender: Male

| Examination   | University  | Institute                        | Year | CPI / % |
|---------------|-------------|----------------------------------|------|---------|
| Graduation    | IIT Bombay  | IIT Bombay                       | 2026 |         |
| Intermediate  | Maharashtra | City Pride Junior College, Nigdi | 2021 |         |
| Matriculation | CBSE        | City Pride School, Nigdi         | 2019 |         |

Pursuing **Minor** in department of **Computer Science and Engineering** at Indian Institute of Technology, Bombay

## SCHOLASTIC ACHIEVEMENTS

- Received the **Best Poster Presentation Award** at RISC Symposium conducted by CSE department at IITB (2025)
- Secured **All India Rank 198** in Kishore Vaigyanik Protsahan Yojana (KVPY) exam among **20,000** aspirants (2021)
- Achieved **All India Rank 495** in **JEE Advanced** | **99.89** percentile in **JEE Main** exam among a **Million** candidates (2021)
- Secured an **All Maharashtra Rank 13** in the **MHT-CET** examination among **0.2 Million** aspirants (2021)

## PUBLICATIONS

S. Prakash et al., "QuArch: A Question-Answering Dataset for AI Agents in Computer Architecture," accepted for publication in the journal **IEEE Computer Architecture Letters (CAL)** (Feb 2025)

## RESEARCH EXPERIENCE

**ArchGPT: LLMs for Computer Architecture** | **Harvard University** (Jan 2024 - Present)  
Prof. Vijay Janapa Reddi | In-semester Research | Edge Computing Lab, Harvard University

- Contributed to development of embeddings and **LLMs** for hardware design to support practitioners and researchers
- Led team to collect **billion-token corpus** of academic papers and industry documentation via **Selenium** web crawling
- Built **AWS**-based pipeline to extract bibliographic and content metadata from academic papers enabling NLP tasks

**Deep Learning for Group Testing: Algorithms and Interpretability** (Apr 2025 - Present)  
Prof. Satish Mulleti and Prof. Ajit Rajwade | **Master's Thesis** | IIT Bombay | To be submitted to ICASSP

- Implemented noise robust Deep Neural Networks for **Group Testing** with accuracy matching the state-of-the-art
- Proposed **model interpretability** approach in Quantitative Group Testing by recovering pools' structure from model
- Compared outcomes with **Maximum Neighborhood** QGT method for varying measurement rates and sparsity levels
- Work to be submitted to the International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2026

**ArchGym: Open-Source Gymnasium for ML-Assisted Architecture Design** (Jun 2023 - Aug 2023)  
Prof. Vijay Janapa Reddi | Research Internship | Edge Computing Lab, Harvard University

- Contributed to ArchGym by building an efficient pipeline to integrate RL algorithms with architecture simulators
- Developed **proxy ML models** for architecture simulators with tunable hyperparameters, enabling fast simulations
- Utilized Deepmind envlogger to record environment-agent interactions; authored comprehensive user documentation

**Tomography Under Unknown Angles and Spatial Shifts** (Aug 2024 - Mar 2025)  
Prof. Satish Mulleti and Prof. Ajit Rajwade | Research Project | IIT Bombay | To be submitted to ICASSP

- Proposed an algorithm for **image reconstruction** in 2D tomography with unknown projection angles and spatial shifts
- Developed **shift-aware initialization** using Laplacian eigenmaps for accurate projection ordering under spatial translations
- Validated on ribosome images, achieving superior recovery over baseline filtered backprojection without shift correction
- Work to be submitted to the International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2026

## PROFESSIONAL EXPERIENCE

**Digital Electronics Intern, Texas Instruments** (May 2024 - July 2024)  
Radar Business Unit | Bengaluru, India

- Computed **Coremark/Dhrystone** performance for multiple radar chip architectures to guide architecture trade-offs
- Analyzed **cache utilization** on QuickTurn emulator; modified cache interactions to simulate subsystem algorithms
- Extracted key performance metrics, demonstrating potential for **30%** lower power and **40%** lower area utilization

## TEACHING AND MENTORSHIP

**Teaching Assistant** | Signal Processing | Prof. Preeti Rao (Aug 2025 - Present)  
Department of Electrical Engineering, IIT Bombay | 100+ students

**Summer of Science Mentor** | Image Processing (May 2025 - July 2025)  
Maths and Physics Club, IIT Bombay | 7 students

KEY PROJECTS

**Human-Feedback-Boosted Generation of Neutrophil Images** (Jan 2024 - May 2024)  
Machine Learning and Statistical Methods in Healthcare | Course Project | Prof. Kshitij Jadhav

- Trained diffusion model from scratch to generate synthetic histopathology images to tackle medical image scarcity
- Designed multi-label **reward model** to guide clinically plausible image generation from expert pathologist feedback
- Improved plausibility from **82% to 90%** via fine-tuning with feature-weighted feedback obtained via **Label Studio**

**Markov Decision Processes and Reinforcement Learning** (Jan 2025 - Apr 2025)  
Foundations of Intelligent and Learning Agents | Course Project | Prof. Shivaram Kalyanakrishnan

- Implemented and evaluated UCB, KL-UCB, Epsilon-Greedy, Thompson Sampling on **multi-armed stochastic bandits**
- Built MDP planner (Howard's Policy Iteration & LP) and encoder/decoder to solve **stochastic gridworld game** optimally
- Designed RL agent with CMA-ES policy search for robust high-speed autonomous driving across unseen racetracks

**Denoising Diffusion Probabilistic Models** (Aug 2023 - Nov 2023)  
Advanced Topics in Machine Learning | Course Project | Prof. Amit Sethi

- Implemented **DDPM** framework with a custom U-Net architecture to synthesize realistic images directly from noise
- Explored and optimized **noise scheduling** strategies (linear vs cosine) to improve sample fidelity and training stability
- Extended the model to color image generation in HSV space, achieving higher quality and diversity of outputs

**Bubble Trouble Game using C++** (Aug 2021 - Nov 2021)  
Computer Programming and Utilization | Course Project | Prof. Parag Chaudhuri

- Developed C++ shooting game with object-oriented programming, parabolic physics, and collision detection systems
- Implemented physics engine featuring gravity, bouncing mechanics, and dynamic bubble splitting on bullet impact
- Built game mechanics with health/score tracking, time limits, and progressive difficulty via bubble size/speed changes

**Stereo Vision and Shape from Shading** (Jan 2025 - Apr 2025)  
Computer Vision | Course Project | Prof. Subhasis Chaudhuri

- Implemented stereo vision algorithms on Middlebury dataset to recover **depth maps** with quantitative evaluation
- Developed shape-from-shading to reconstruct **3D geometry** from single image under noise and illumination variations

**Embedded DSP System Design** (Jan 2025 - Apr 2025)  
Digital Signal Processing Software and Hardware Lab | Course Project | Prof. Preeti Rao

- Optimized real-time FIR and IIR filter implementation on TI ezDSP5515 using C and Assembly with **circular buffers**
- Accelerated spectrum analysis by using DMA with **ping-pong buffers** to feed a dedicated FFT hardware accelerator
- Designed a hardware-based **DTMF decoder** and developed multirate filters for DSP systems in GNU Radio

**FPGA-Based RRAM Characterization Instrument** (Jan 2024 - May 2024)  
Electronic Design Lab | Course Project | Prof. Siddharth Tallur

- Led design of **analog frontend** (amplifiers, ADC/DAC interface) for FPGA-based RRAM characterization system
- Developed Python GUI to **interface FPGA with laptop**, enabling real-time control, data acquisition, and visualization
- Delivered end-to-end hardware pipeline, from schematic and PCB design to prototype testing, achieving < 5% error

TECHNICAL SKILLS

**Softwares/Languages** Python, C, C++, MATLAB, VHDL, Assembly, GNURadio, Quartus Prime, Keil  $\mu$ Vision  
**Libraries** PyTorch, NumPy, Scikit-Learn, Pandas, Matplotlib SciPy, OpenCV, Selenium, OpenAI Gym  
**Miscellaneous** OpenMP, MPI, CUDA C, L<sup>A</sup>T<sub>E</sub>X, NgSPICE, Google Cloud Platform, Amazon Web Services

RELEVANT COURSEWORK

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine Learning and Data Science | Advanced Topics in Machine Learning, Machine Learning for Remote Sensing, Foundations of Intelligent and Learning Agents, Programming for Data Science |
| Computer Science                  | Advanced Image Processing, Medical Image Computing, Operating Systems, Data Structures & Algorithms, Computer Networks, Advanced Computer Architecture |
| Electrical                        | Digital Signal Processing (Course and Lab), Control Systems, VLSI Design                                                                               |

EXTRACURRICULAR ACTIVITIES

- Volunteered for management and leadership for various treks conducted by the Adventure Club at IIT Bombay
- Successfully completed one year of professional training under the National Sports Organization (NSO)
- Attained A2 level proficiency in the German examinations conducted by the Goethe Institut, Max Mueller Bhavan
- Was member of the school Harmonium team and completed two levels of the Akhil Bharatiya Gandharva exams
- Won first prize under the autonomous category in the ROBOTRICKS competition conducted by ThinkLabs
- Was part of the school core Basketball team for five years and participated in various competitions